Seonghyun Jin

Integrated M.S./Ph.D. Student, KAIST Graduate School of AI

jinotter3@kaist.ac.kr jinotter@gmail.com jinotter3.github.io github.com/jinotter3 Google Scholar

Research Interests

Reconstructing and generating the real 3D world; streaming 3D reconstruction; geometric perception; Geometric Positional encoding; camera-controlled video generation.

Education

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2025 – Present

Daejeon, Republic of Korea

Integrated M.S./Ph.D. in Artificial Intelligence

- Advised by Prof. Jong Chul Ye at BISPL.

Seoul National University (SNU) Mar. 2020 – Feb. 2025

Seoul, Republic of Korea

B.S. in Computer Science; B.A. in Economics, College of Liberal Studies

- GPA: 3.68/4.3, *Cum Laude*.

Publications

P: preprint, C: conference

[P2] **FILT3R: Latent State Adaptive Kalman Filter for Streaming 3D Reconstruction**
Seonghyun Jin, Jong Chul Ye

Under review. arXiv:2603.18493.
PDF Code Project

[P1] **CRePE: Curved Ray Expectation Positional Encoding for Unified-Camera-Controlled Video Generation**
Seonghyun Jin, Youngmin Kim, Sunwoo Park, Jong Chul Ye

Under review. Preprint in preparation, 2026.
arXiv link to be added after upload.

[C1] **AI-Based Mental Health Assessment for Adolescents Using Their Daily Digital Activities**
Do Hyung Kim, Joonsung Lee, Taehwi Lee, Soeun Baek, **Seonghyun Jin**, HaEun Yoo, Youngeun Cho, Seonghyeon Park, Kwangsu Cho, Chang-Gun Lee

IEEE DSAA, 2024.
PDF 6 citations

Academic Experience

RUBIS Lab, Seoul National University Sep. 2022 – Jun. 2024

Seoul, Republic of Korea

Undergraduate Research Intern

- Researched digital phenotyping for adolescent mental health assessment from daily digital activity data.
- Performed feature engineering on human-stroke/scribble data for downstream AI-based mental health prediction.
- Built a React-based data collection application for a longitudinal user study with approximately 300 adolescent participants over 2 months.
- Contributed to an IEEE DSAA 2024 publication as co-author.

Languages

Korean (native); English (fluent, TEPS 538, TOEIC 980).

Technical Skills

Programming Python, C, Dart, PyTorch, Git, LaTeX
Research 3D reconstruction, computer vision, deep learning, experimental evaluation